

– Supplementary Material –

A. Datasets

We conduct experiments on two co-speech motion datasets, BEAT2 and SHOW. BEAT2 is used for model training and in-domain evaluation, while SHOW is used exclusively for zero-shot cross-dataset evaluation. For text-conditioned methods, transcripts and word-level timestamps are automatically extracted from SHOW audio using a frozen speech-to-text model and shared across all methods. This setting evaluates both in-domain generation performance and robustness to unseen speakers, motion distributions, and recording conditions without target-domain adaptation.

1) *BEAT2*. BEAT2 [1] is a large-scale multimodal dataset for holistic co-speech motion generation, providing temporally aligned speech, text, and SMPL-X motion covering facial expressions, body motion, and hand articulation. We use its English subset and follow the official 85%:7.5%:7.5% training, validation, and test split. Motions are represented at 30 FPS using 330-dimensional 6D joint rotations, 3D root translations, four contact labels, and 100-dimensional facial expression coefficients, while audio is sampled at 16 kHz. Training clips contain 64 frames with a stride of 20, with the first four frames used as seed motion. We report both *one-speaker* and *all-speaker* results and use BEAT2 for all ablation and representation analyses unless otherwise stated.

2) *SHOW*. SHOW [2] is an in-the-wild holistic co-speech motion dataset reconstructed from monocular talking videos and represented with SMPL-X. Compared with the controlled BEAT2 setting, SHOW exhibits substantial shifts in speaker identity, gesture style, motion statistics, and recording conditions. We therefore use its official test split exclusively for zero-shot cross-dataset evaluation. Since SHOW does not provide the textual annotations required by text-conditioned methods, transcripts and word-level timestamps are automatically obtained from the test audio using a frozen speech-to-text system. Detailed preprocessing and inference protocols are described in the following subsection.

B. Zero-Shot Cross-Dataset Evaluation on SHOW

1) *Evaluation split*. We follow the TalkSHOW/SemTalk protocol for SHOW and use the corresponding test split for evaluation, including all valid test sequences without performance-based sample selection. All evaluated models are trained on the BEAT2 English all-speaker split and directly applied to SHOW; no SHOW data are used for training, fine-tuning, model selection, or parameter updates.

2) *Speech transcription*. For text-conditioned methods, transcripts and word-level timestamps are obtained once from SHOW test audio using WhisperX with the `faster-whisper-large-v3` model, with English specified as the transcription language and the standard WhisperX word-level alignment procedure. No manual correction is

performed. The identical ASR outputs are shared by all text-conditioned methods, while audio-only methods retain their original input protocol.

3) *Zero-shot inference*. All SHOW experiments use models trained on the BEAT2 English all-speaker split. Since SHOW speakers are unseen during BEAT2 training, we use a fixed speaker condition obtained by averaging the learned speaker embeddings of all BEAT2 training speakers. The resulting mean embedding is shared by all SHOW sequences and remains fixed throughout inference, thereby avoiding any speaker-specific adaptation or identity matching using SHOW data. For methods requiring motion history, only the first H ground-truth frames are used for initialization; all subsequent windows rely exclusively on previously generated motion, and initialization frames are excluded from metric computation.

4) *Motion preprocessing and evaluation*. SHOW motions are converted to the same SMPL-X representation and coordinate convention used for BEAT2. The same preprocessing and evaluation pipeline is used for all methods. We report FGD, BC, DIV, MSE, and LVD. SRGR is omitted because SHOW does not provide the gesture-level semantic relevance annotations required by this metric.

C. Additional Implementation Details

All experiments are conducted on a single NVIDIA A100 GPU. Following the protocol of EMAGE [1], motion sequences are sampled at 30 FPS and represented by 330-dimensional 6D SMPL-X joint rotations, 3D root translations, four foot-contact labels, and 100-dimensional FLAME-compatible facial expression coefficients. Audio is sampled at 16 kHz. For speech conditioning, we use onset and amplitude features, 1,024-dimensional HuBERT representations, 300-dimensional fastText word embeddings, and CLIP ViT-B/32 sentence embeddings.

Each training sample contains 64 motion frames with a stride of 20, where the first four frames are used as seed motion. The body backbone has a hidden dimension of 768, and each regional RVQ motion latent has 256 dimensions. The speech-to-plan predictor consists of a two-layer Transformer encoder with a feature dimension of 128, four attention heads, a feed-forward dimension of 512, and a dropout rate of 0.1.

The future motion-plan tokenizer contains two residual MLP blocks and a codebook with $K = 64$ entries of dimension $D_p = 128$. We use a future horizon of $H = 2$ motion tokens and an offset of $o = 1$, corresponding to the subsequent eight motion frames (approximately 0.267 s at 30 FPS). The VQ commitment weight is set to 0.25.

Training is performed in two stages. In Stage I, the future motion-plan tokenizer is pretrained for 100 epochs using Adam with $\beta_1 = 0.5$ and $\beta_2 = 0.999$, a batch size of 128, and a learning rate of 2×10^{-4} . After pretraining, the future-window encoder, decoder, and plan codebook are frozen. In



Fig. 1. Qualitative facial-motion comparison. Red dashed boxes mark temporal inconsistencies. PlanTalk shows smoother dynamics while retaining SemTalk’s speech-conditioned facial pathway without plan modulation; thus, the results reflect overall system quality rather than a direct planning benefit.

Stage II, the speech-to-plan predictor, event-aware semantic gate, hierarchical plan-conditioned modules, and motion generator are jointly trained for 200 epochs with a batch size of 64 and an effective learning rate of 5×10^{-5} . We use no weight decay and clip the gradient norm to 0.99.

During training, discrete plan selection is implemented using straight-through Gumbel-Softmax with a temperature of 1.0. At inference, the plan index with the maximum prediction logit is selected deterministically. To stabilize optimization, the plan-related objective is disabled during the first 10 epochs and linearly introduced over the following 10 epochs.

We set $\lambda_p = 0.2$, $\lambda_{\text{align}} = 0.1$, $\lambda_{\text{ctr}} = 0.1$, and $\lambda_{\text{event}} = 0.5$. The contrastive temperature is set to 0.1, and all experiments use a random seed of 43 unless otherwise specified.

D. Qualitative Comparison of Facial Motion

Fig. 1 further compares the facial motion generated by different methods on the same speech segment. Although existing methods can produce plausible facial expressions at individual frames, they sometimes exhibit temporal inconsistencies, as highlighted by the red dashed boxes. In particular, GlobalDiff, GestureLSM, StreamTalk, and SemTalk show occasional abrupt changes in head orientation, facial expression, or mouth configuration across neighboring frames, resulting in less coherent facial dynamics. In contrast, PlanTalk produces smoother transitions while maintaining a more stable facial structure and temporally consistent speech-related mouth motion. The resulting facial dynamics also show closer qualitative consistency with the ground-truth sequence.

We note that the facial stream in PlanTalk is directly conditioned on speech and is not modulated by the predicted future plan. Therefore, this comparison is intended to demonstrate the overall facial-generation quality of the full system, rather than to attribute facial improvements directly to the proposed future-planning mechanism.

TABLE I
EFFECT OF FUTURE-MOTION CONTENT AND TEMPORAL ORDERING. ALL VARIANTS SHARE THE SAME DOWNSTREAM ARCHITECTURE AND TRAINING PROTOCOL.

Representation	FGD↓	BC↑	DIV↑	SRGR↑	Event F1↑
Future Endpoint Block	4.168	0.803	13.68	0.433	0.731
Endpoint + Energy	4.103	0.807	13.75	0.436	0.746
Order-Invariant Future Set	4.027	0.812	13.84	0.438	0.758
Ordered Future Sequence	3.812	0.817	13.95	0.442	0.784

E. Effect of Temporal Ordering in Future Plans

To determine whether the proposed future plan benefits from temporal ordering, we compare representations containing progressively richer information about the same future interval. All variants use the same backbone, plan dimensionality, code-book size, planner, motion generator, reconstruction target, and training protocol, differing only in the information provided to the plan tokenizer.

Let $\{u_{t+o+j}\}_{j=0}^{H-1}$ denote the local motion representations within the future window. *Future Endpoint Block* retains only its final L -frame block:

$$W_t^{\text{end}} = u_{t+o+H-1}. \quad (1)$$

Endpoint + Energy additionally includes the average motion energy $\bar{E}_t$ over the complete window:

$$W_t^{\text{end+energy}} = [u_{t+o+H-1}; \bar{E}_t]. \quad (2)$$

Order-Invariant Future Set retains all future blocks but removes their relative temporal positions using a shared projection and symmetric aggregation:

$$h_t^{\text{set}} = \rho \left(\frac{1}{H} \sum_{j=0}^{H-1} \phi(u_{t+o+j}) \right). \quad (3)$$

This construction preserves the constituent motion content while being invariant to their permutation. In contrast, *Ordered Future Sequence* encodes the same blocks in their original order:

$$h_t^{\text{ord}} = E_m(\text{Vec}(u_{t+o}, \dots, u_{t+o+H-1})). \quad (4)$$

Both variants reconstruct the same canonically ordered raw-motion window, isolating the contribution of within-window temporal order.

As shown in Table I, augmenting the future endpoint with motion energy yields consistent numerical improvements across all metrics, reducing FGD from 4.168 to 4.103 while increasing BC, DIV, SRGR, and Event F1. Retaining all future blocks through order-invariant aggregation further improves FGD to 4.027, BC to 0.812, DIV to 13.84, SRGR to 0.438, and Event F1 to 0.758, indicating that window-level motion content provides richer guidance than an endpoint or aggregate motion statistics alone. Restoring temporal order achieves the best overall performance, with an FGD of 3.812, BC of 0.817, DIV of 13.95, SRGR of 0.442, and Event F1 of 0.784. Since the invariant and ordered variants contain the same future blocks, their difference supports the usefulness of within-window temporal ordering for short-horizon motion organization and semantic-event activation.

TABLE II
LINEAR PROBING OF ORACLE AND INFERENCE-TIME REPRESENTATIONS.
ALL REPRESENTATIONS ARE FROZEN, AND EACH ROW USES THE SAME
PROBING PROTOCOL.

Representation	Cur. Part F1↑	Fut. Part F1↑	Fut. Energy R^2 ↑	Temp. Evol. F1↑
Frozen Speech Feature	0.575	0.629	0.415	0.551
Current Motion Token	0.792	0.558	0.327	0.486
Speech-Predicted Future Plan	0.582	0.697	0.544	0.651
Oracle Continuous Future Feature	0.624	0.781	0.684	0.746
Oracle Discrete Future Plan	0.603	<u>0.754</u>	<u>0.631</u>	<u>0.718</u>

F. Understanding the Learned Future Plans

We examine whether the future structure encoded by the motion tokenizer remains accessible in the speech-predicted plan used at inference. We compare the frozen speech feature s_t , the current motion token, the oracle continuous future feature h_t^m , the oracle discrete plan p_t^m , and the predicted plan

$$\hat{k}_t = \arg \max_k P(s)_{t,k}, \quad \hat{p}_t^m = c_{\hat{k}_t}, \quad (5)$$

which is exactly the condition passed to the generator and uses no future motion.

We probe current and future body-part activation, future motion energy, and future temporal evolution. Regional energy is defined as the mean squared joint velocity within the corresponding block or future window. Region r is active when its energy exceeds the training-set median $\tau_r = \text{median}_{\text{train}}(e_r)$. Temporal evolution is labeled as increasing, decreasing, or stable according to $\Delta e_t = e_{t+o+1} - e_{t+o}$. We set δ to the 33.3rd percentile of $|\Delta e_t|$ on the training split and use $\pm\delta$ as thresholds. All thresholds are estimated from training data and fixed for validation and testing.

All representations and planners are frozen. We use the same utterance-level splits as the main experiments and standardize features with training-set statistics. Classification probes are single affine layers trained with class-balanced cross-entropy, while future energy uses ridge regression. The regularization coefficient is selected on the validation set from $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$. We report macro-F1 for classification and R^2 for regression, using only valid positions with $m_t = 1$.

As shown in Table II, the current motion token achieves the highest current-part activation F1 of 0.792 but provides limited predictability of future attributes. The frozen speech feature already outperforms the current token on future-part activation (0.629 vs. 0.558), future energy (0.415 vs. 0.327), and temporal evolution (0.551 vs. 0.486), confirming that speech contains cues about upcoming motion.

The speech-predicted plan further improves these scores to 0.697, 0.544, and 0.651, corresponding to gains of 0.068 in future-part F1, 0.129 in future-energy R^2 , and 0.100 in temporal-evolution F1 over the frozen speech feature. This indicates that the planner reorganizes speech cues into a representation where future body participation, motion magnitude, and within-window evolution are more linearly accessible.

The oracle continuous feature achieves the strongest future-oriented scores of 0.781, 0.684, and 0.746. Quantization reduces them moderately to 0.754, 0.631, and 0.718, showing

that the discrete bottleneck retains most of the future structure. The remaining gaps of 0.057, 0.087, and 0.067 between the speech-predicted and oracle discrete plans quantify speech-to-plan prediction loss. Since $\hat{p}_t^m$ is a deterministic transformation of speech, these gains reflect improved linear accessibility of future-motion factors rather than new information absent from speech.

We further contextualize code-prediction accuracy using codebook utilization and frequency baselines. Among the 64 codes, 59 are used, with perplexity $\exp(-\sum_k f_k \log(f_k + \epsilon)) = 39.6$, where f_k denotes the training-set code frequency. Always predicting the most frequent code gives Prior@1 of 0.061, while selecting the five most frequent codes gives Prior@5 of 0.226, compared with planner Acc@1/Acc@5 of 0.536/0.798 and macro-F1 of 0.468. These statistics use the shared frozen codebook; accuracies from separately trained codebooks are not directly comparable.

G. Temporal Motion Evolution

To characterize the temporal evolution of body motion, we compute the motion energy at frame t as the average frame-wise joint displacement:

$$E_t = \frac{1}{J} \sum_{j=1}^J \|\mathbf{x}_{t,j} - \mathbf{x}_{t-1,j}\|_2, \quad (6)$$

where J denotes the number of body joints and $\mathbf{x}_{t,j}$ is the 3D position of joint j at frame t . A larger E_t indicates stronger instantaneous body motion, while its temporal profile reflects how motion intensity evolves throughout the speech segment.

Fig. 2 visualizes the motion-energy evolution over a representative 10-second utterance containing five semantically salient speech events: *opened* (1.33 s), *shocked* (4.05 s), *everyone* (5.64 s), *surprise* (7.65 s), and *today* (9.14 s). The ground-truth motion exhibits pronounced energy increases around these semantic landmarks, forming a sequence of smoothly connected high- and low-motion intervals. PlanTalk closely follows this temporal structure, with major energy peaks occurring near the same semantic events and smooth transitions between successive responses, although some peak magnitudes remain slightly lower than those of the ground truth.

In comparison, StreamTalk captures several high-motion intervals but shows noticeable temporal shifts and additional local variations around some semantic events. SemTalk produces more localized responses with sharper energy concentration and weaker activity between successive events, while GlobalDiff exhibits lower-amplitude and frequent local fluctuations throughout the sequence. Overall, PlanTalk better preserves the coarse temporal organization of the reference motion across multiple semantic events, providing qualitative evidence consistent with the role of future-window planning in organizing upcoming motion dynamics.

H. User Study Details

To complement the objective evaluation, we conduct a perceptual user study to assess the visual quality and speech-motion consistency of the generated motions. Following gen-

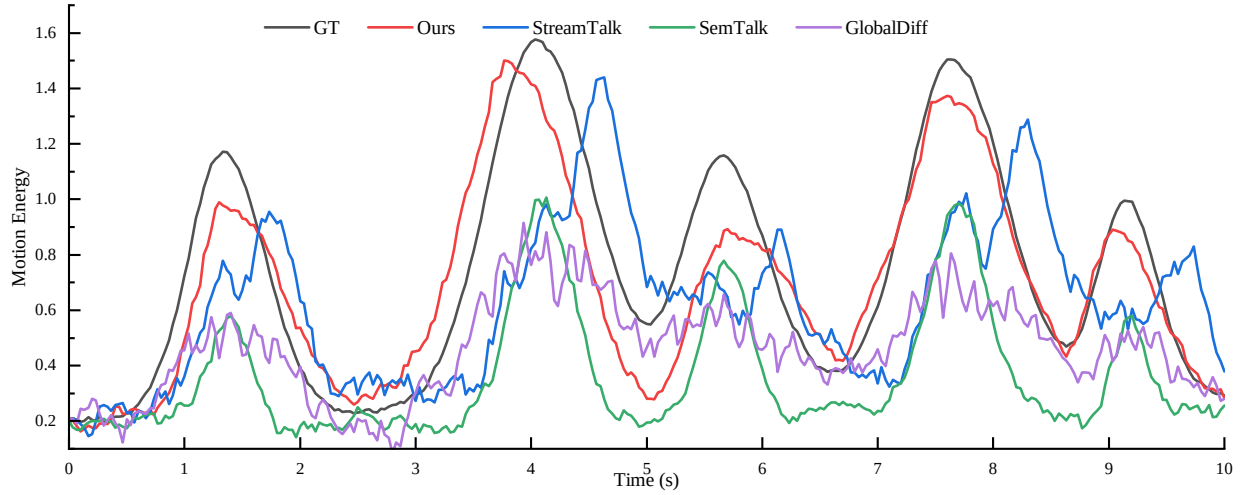


Fig. 2. Motion-energy evolution over a representative 10-second utterance containing five salient semantic events. PlanTalk more closely follows the ground-truth temporal profile across successive events, while competing methods show temporal shifts, localized responses, or frequent fluctuations.

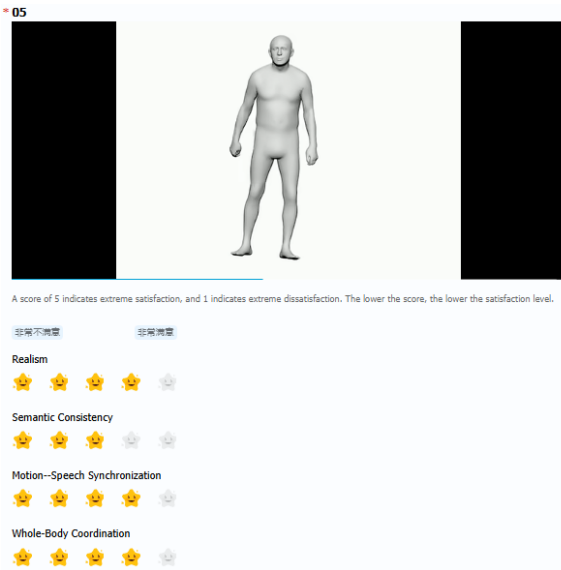


Fig. 3. Screenshot of user study website.

eral subjective assessment principles established by the International Telecommunication Union (ITU) [3], we recruit 30 participants to evaluate four methods: SemTalk, GlobalDiff, StreamTalk, and PlanTalk. The same ten speech sequences are rendered by all four methods, resulting in 40 matched videos in total. Every participant evaluates all matched outputs in randomized order. Therefore, for each method and each perceptual criterion, the study collects $30 \times 10 = 300$ individual ratings.

Participants evaluate each sequence using a five-point Mean Opinion Score (MOS) scale, where a higher score indicates better perceptual quality. Four complementary criteria are considered: *Realism*, measuring the overall naturalness and plausibility of the generated motion; *Semantic Consistency*, assessing whether the gestures are appropriate for the corresponding speech content; *Motion-Speech Synchronization*, evaluating the temporal alignment between speech and body motion; and *Whole-Body Coordination*, measuring the coherence of motion across the upper body, hands, torso, and lower body. The questionnaire interface used in the study is

illustrated in Fig. 3.

Prior to evaluation, participants are provided with instructions explaining the four criteria and the five-point scoring scheme. For each method and criterion, the MOS is computed over all 300 individual ratings. For visualization, ratings are first averaged over the ten matched sequences for each participant, yielding 30 participant-level mean scores; error bars report the standard deviation across these scores. Because the design is balanced, the mean of the participant-level scores is identical to the MOS obtained from all 300 ratings. To assess the consistency of subjective judgments across raters, inter-rater reliability is measured using Krippendorff's alpha, yielding $\alpha = 0.78$.

Because each participant evaluates the same set of ten speech sequences for all four methods, the resulting ratings are not treated as 300 independent observations in statistical significance testing. Instead, for each participant, method, and perceptual criterion, we first average the ratings over the ten matched speech sequences, yielding one participant-level score per method and criterion. Pairwise comparisons between PlanTalk and each competing method are then performed using the Wilcoxon signed-rank test on the resulting 30 paired participant-level scores. This preserves the within-participant pairing while avoiding inflation of the effective sample size caused by treating repeated ratings from the same participant as independent observations. Statistical significance is evaluated at $p < 0.01$.

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